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ASSET PRICING & PORTFOLIO THEORY

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Abstract

This paper's idea is to combine portfolio strategies and thereby deduct the resulting demand for each asset from a chosen portfolio of assets. The deduction of demand allows the model to formulate new weights for the assets that perform well in back testing. The high return and the high volatility suggest some bubble riding effects of demand estimation. However, depending on the interpretation of fundamental value in portfolio strategies, the mechanisms of this paper's portfolio strategy can be seen as fundamental value combining instead of bubble riding.

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1. Introduction

This paper uses the data from the group work of Marta Rodrigues, Leonor R. Pereira, Bernardo Manarte and me (Leopold Kaiser). This data consists of 22 assets and nine different portfolio strategies that were applied and backtested on the assets returns. The ninth portfolio strategy in the data combines the weights of the previous eight portfolio strategies, labelled "Global Weight Portfolio" (GWP). The GWP construction starts with a matrix which elements are the weights that the existing eight portfolios allocated to the 22 assets. Those eight portfolios are equally weighted as a starting point, in this analysis that corresponds to a $\frac{1}{8}$ share for each of the existing eight portfolios. This "Equal Global Weight Portfolio" (EGWP) performs well in back testing with a Sharp Ratio of 0.743.

This paper aims to find a more sophisticated approach to find the weights of the portfolios, instead of simply equally weighting them. The main idea is that the weights of the prevailing eight portfolios and their weight allocation for each asset corresponds to the assets demand, with the assumption that all the demand results from the prevailing portfolio strategies.

2. Approach outline

The model's aim is to find the corresponding weights of the eight portfolios that resulted in the assets demand. The model's proxy for demand is the normalized return, with the background that in a system with all demand coming from the eight preexisting portfolio strategies and equal supply, the return of an asset relative to the total return of all assets in the portfolio corresponds to the assets demand. Given this background, the portfolio strategy that finds p_j in the following outlined way, shall be named "Demand Matrix Portfolio" (DMP). Mathematically the logical chain of equalities in this model's framework can be expressed as follows:

$$\frac{\mathrm{R_i}}{\sum_{j=1}^{n} R_j} = D_i = \sum_{j=1}^{n} w_{ij} p_j \qquad (1)$$

In equation (1), $\mathrm{R_i}$ corresponds to the return of an asset i, $\sum_{j=1}^{n} R_j$ is the sum of the returns of all assets in the portfolio (in our analysis the sum of the return of 22 assets), D_i corresponds to the demand of the asset i and $\sum_{j=1}^{n} w_{ij} p_j$ corresponds to the demand matrix with w_{ij} being the weights found for each asset i by the different portfolio strategies and p_j being the weights of the portfolio strategies themselves (in our case eight different weights for eight different portfolios). The goal of the model is to find the vector p_j, such that equation (1) is satisfied for every asset. In the following, three different approaches are outlined to reach this goal, and one is being chosen.

2.1 Weight combinations

The first approach is to create every possible combination of weights in the vector p_j and looping them through equation (1) until the equality holds or the closest value to the equality is found. The issue with this approach is that it is a) computational expensive and b) might sometimes be misleading since it could find a weights combination by accident that satisfies (or almost satisfies) equation (1) but doesn't correspond to the actual weight/ demand allocation. When still choosing this approach there exists a trade-off between different weight steps, number of simulated portfolios and Random Access Memory (RAM) capacity. Since the smaller the steps between the weights and the higher the number of calculated portfolio strategies, the higher will be the number of combinations necessary to compute every possible combination and therefore the RAM capacity. In our analysis we have eight portfolios, so if we take steps of 0.1 and build the cartesian product for the right part of equation (1), the number of calculations would correspond to $\left(\frac{1}{0.1}\right)^8 = 100\,000\,000$. There are multiple ways to address this issue, like using larger weight steps or expanding the RAM in the programming language or use more efficient data storage properties like the "data.table" in R or the "yield" statement in Python, or use cloud computing for example with Google Colab. However, given the two flaws a) and b), for this model we do not take the weight combinations approach to find p_j .

2.2 Analytical Solution

The analytical solution would be a feasible solution to find p_j. However, when solving equation (1) for the vector of p_j in the demand matrix we encounter an overidentified system of equations resulting in misleading derived weights, since it's not always possible to find an exact solution for all equations due to the possibility of inconsistencies between them. The analytical solution could only solve for the exact p_j when the number of equations equal the number of variables to solve for, resulting in a square system of equations. In our case it would be necessary to have the same number of assets as preexisting portfolio strategies. Since the rows in the demand matrix correspond to the number of assets and they must be solved for the vector of p_j.

In our analysis, we have 22 equations (from 22 assets) but only eight variables to solve for (number of portfolio strategies), given this overidentified system, a least square approach is more feasible.

2.3 Regression

Following on the previous argumentation, a least square approach seems to be the most feasible for this model's framework. To take the non-normally distributed data into account, a General Least Square (GLS) method was chosen. However, the Ordinary Least Square (OLS) method was tested and resulted in the same coefficients, significance and explanatory power as the GLS.

After running the regression, the vector of coefficients ß is normalized to build portfolio weights after accounting for no short selling and excluding negative coefficients from the weight calculation to find p_j. The normalization takes place as before with the returns, by dividing the individual coefficient of a portfolio strategy by the sum of all positive portfolio strategies coefficients: $\frac{\beta_i}{\sum_{j=1}^{n} \beta_j} = p_i$. After normalization p_j has the classical weight attribute: $\sum_{j=1}^{n} p_j = 1$

To run the regression, equation (1) must be modified:

$$\frac{\mathrm{R_i}}{\sum_{j=1}^{n} R_j} = D_i = \beta_0 + w_i^T ß + u_i$$

As in equation (1), $\frac{\mathrm{R_i}}{\sum_{j=1}^{n} R_j}$ represents the vector of relative returns. In this regression methodology β_0 represents the intercept term, which might represent a systematic bias or base level of demand not captured by the portfolio weights. w_i^T represents the transposed vector of portfolio weights for asset i found by the preexisting portfolio strategies, in our analysis eight. u_i is the error term. $w_i^T ß$ can already by interpreted as undimensional demand vector,

undimensional because ß is yet not normalized. The dimension in this Case is [0, 1] or $\sum_{j=1}^{n} p_j = 1$.

As already mentioned in the introduction of this paper, this model assumes that the demand is fully captured by the existing portfolio strategies. Given this assumption the intercept should be zero and we can modify the regression equation as follows:

$$\frac{\mathrm{R_i}}{\sum_{j=1}^{n} R_j} = D_i = w_i^T ß + u_i$$

$$u_i = D_i - w_i^T ß$$

The regression then minimizes the sum of squared residuals (SSR), corresponding in this model to:

$$SSR = \sum_{j=1}^{n} u_i^2$$

The GLS method will find the coefficient vector ß, such that SSR is minimized only leaving unfitted values as error terms. The error terms u_i can be interpreted as the component of demand not captured by the portfolio weights.

The results in this regression are mostly insignificant, possibly due to a low number of observations in the matrix. The significance might increase with more assets in the portfolios. And since there are more portfolio strategies out there than just eight that determine the demand for one asset, supply kept constant, this regression might include omitted variable bias. So, adding more portfolios to the model and therefore to the regression method could improve the fit in the method and increase the best linear unbiased estimator properties of the coefficients.

3. Results

In the back testing of the "Demand Matrix Portfolio" (DMP), the weights are recalculated and recalibrated every 63 days. The DMP strategy is applied on random samples of 10 assets on 100 datasets for two consecutive years, randomly drawn between 01.01.2010 until 08.11.2024. To compare the performance metrics of the DMP, it is compared to the S&P 500 Index and the "Equal Global Weight Portfolio" (EGWP).

Metric	DMP	EGWP[1]	Index
Sharpe Ratio	1.959	0.743	0.698
Max Drawdown	0.79	0.148	0.142
Annual Return (%)	6.919	0.103	0.116
Annual Volatility	3.636	0.143	0.149
Sortino Ratio	6.345	1.051	0.965

[1] Values taken from group analysis

Downside Deviation	1.151	0.099	0.106
Sterling Ratio	8.754	0.785	0.61
Omega Ratio	1.912	1.137	1.134
VaR (0.95)	0.146	0.014	0.016
CVaR (0.95)	0.241	0.02	0.023
Rebalancing Period	63	63	441
Turnover	0.303	0.006	0

Table 1: Performance comparison of the Demand Matrix Portfolio (DMP), Equal Global Weight Portfolio (EGWP) and the S&P 500 Index (Index)

Table 1 shows performance measured as either a) plain return / return relative to risk or b) risk. The conditional colour formatting scale reaches from green to red, corresponding to a scale from advantageous for investors to disadvantages for investors. The middle of the spectrum is displayed as yellow.

For the DMP, it becomes visible that the performance measures a) are strongly positive for investors in comparison to EGWP and Index, whereas performance measures b) are relatively negative for investors.

When inspecting the colour distribution between EGWP and Index, one can see that the index mostly performs better in pure performance measures, whereas in terms of performance in relation to risk the EGWP performs better than the index. Interestingly, the EGWP performs better in terms of volatility measures (except from Max Drawdown), this can suggest that the EGWP methodology inhabits stronger risk diversifying properties than the index.

Under the category performance measures a) fall: "Annual Return (%)", Sharpe Ratio, Sortino Ratio, Sterling Ratio and Omega Ratio. Performance measures b) comprise: Max Drawdown, Annual Volatility, Downside Deviation, VaR (0.95) and CVaR (0.95). Since the performance measures were already described in the group part, this paper will refrain from going into detailed descriptions of the individual performance measures here.

4. Discussion

A Sharp Ratio of close to two might indicate that the economic intuition of this model and its application work well to forecast assets demand. The high volatility can be a result from uncaptured portfolio strategies that resemble an omitted variable bias in the regression, assuming that this models assumptions about the relationship between demand, return and portfolio strategies hold true. Since this model aims to predict demand and allocates weights to assets accordingly, one might describe this behavior as bubble riding, if the expected demand increase only results from behavior that doesn't consider fundamental value. However, fundamental value in the context of portfolio seems to be hard to determine since the objective generally is not to calculate the fundamental value of an asset, but to combine those assets optimally to gain a desired portfolio property. In the case of the Maximum Sharp Ratio portfolio (MSRP) for example, one might consider the systematic risk as fundamental value (one has to pay for with the risk premium in the CAPM framework). How does this concept relate to this

paper's model, that tries to estimate how much demand of one or multiple assets result from the MSRP and other portfolios to find the accumulated demand for one or more assets? – If every portfolio strategy has a fundamental value according to its optimization objective, then this model's strategy can be interpreted as the weighted effect of each portfolio's fundamental value. Given the methodology of this paper, the portfolio's fundamental value would be reflected in the return, when thinking about the prior interpretation. The prior interpretation might seem far-fetched but given the overall abstractness of portfolio strategies it can be feasible.

5. Conclusion

This paper's model performs extraordinary well in back testing when it comes to performance measured in terms of return or return in relation to risk. However, risk performance measures show the strong speculative properties of the DMP approach. When economic intuition of this paper is applicable, then risk measures of the DMP strategy should lower with increasing considered portfolio strategies and assets, since then the properties of the regression framework can be improved. Alternatively, risk measures should decrease when the number of considered portfolio strategies is exactly equal to the number of assets, since then an analytical solution can be derived. Furthermore, when following the economic intuition behind the DMP portfolio, its implementation can be interpreted as fundamental value combining instead of bubble riding.